

Autocovariance

$$\gamma_j = \mathbb{E}[(y_t - \mu)(y_{t-j} - \mu)] \quad (1)$$

Autocorrelation

$$\rho_j = \frac{\gamma_j}{\gamma_0} = \frac{\gamma_j}{\text{Var}[y]} \quad (2)$$

AR(p): ACF $\rightarrow$ gradual decay

MA(q): ACF $\rightarrow$ cutoff after a few lags

Partial autocorrelation

The p -th partial autocorrelation is the p -th coefficient of a linear regression of y_t on its lags up to p :

$$y_t = \hat{\alpha} + \hat{\beta}_1 y_{t-1} + \hat{\beta}_2 y_{t-2} + \dots + \hat{\beta}_p y_{t-p} + \varepsilon_t \quad (3)$$

$$PACF(p) = \hat{\beta}_p$$

AR(p): PACF $\rightarrow$ abrupt dropoff in PACF after lag p

MA(q): PACF $\rightarrow$ gradual decay towards zero in PACF

Source: IMF Course: MFx – Macroeconomic Forecasting

Autocovariance

$$\gamma_j = \mathbb{E}[(y_t - \mu)(y_{t-j} - \mu)] \quad (4)$$

Autocorrelation

$$\rho_j = \frac{\gamma_j}{\gamma_0} = \frac{\gamma_j}{\text{Var}[y]} \quad (5)$$

AR(p): ACF $\rightarrow$ gradual decay

MA(q): ACF $\rightarrow$ cutoff after a few lags

Partial autocorrelation

The p -th partial autocorrelation is the p -th coefficient of a linear regression of y_t on its lags up to p :

$$y_t = \hat{\alpha} + \hat{\beta}_1 y_{t-1} + \hat{\beta}_2 y_{t-2} + \dots + \hat{\beta}_p y_{t-p} + \varepsilon_t \quad (6)$$

$$PACF(p) = \hat{\beta}_p$$

AR(p): PACF $\rightarrow$ abrupt dropoff in PACF after lag p

MA(q): PACF $\rightarrow$ gradual decay towards zero in PACF

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